Graph-Smooth Sparse Orthogonal PCA via Alternating Manifold and Graph-Coupled Proximal Updates

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Abstract

We study sparse, graph-smooth, explicitly orthogonal multi-component PCA through a split-variable formulation that couples an orthogonal loading matrix with a sparse graph-regularized proxy. We propose a concrete alternating solver consisting of a Stiefel manifold update for the orthogonal factor and a proximal-gradient update for the sparse graph-smooth factor. We evaluate the method on synthetic graph families using shared evaluation metrics (support recovery, graph smoothness on the true graph, and shared explained variance). In our synthetic experiments, graph regularization yields smoother solutions on the true graph, while its effect on support recovery is topology-dependent: on chain graphs the method is competitive on recovery and smoother on the true graph, whereas on SBM graphs stronger regularization induces a recovery–smoothness tradeoff and light regularization can improve smoothness with limited recovery loss. The paper is synthetic-first and modest in scope; broader theoretical guarantees and real-data validation are deferred.

1 Introduction

Principal component analysis (PCA) is a foundational technique for dimensionality reduction, exploratory analysis, and low-dimensional representation learning. Its appeal stems from its simplicity and its optimality for variance maximization under orthogonality constraints. However, classical PCA typically produces dense loading vectors, so that each principal component depends on nearly all input variables. In modern high-dimensional applications such as genomics, neuroimaging, and structured sensing, these dense loadings can be difficult to interpret and of limited use for scientific discovery or feature selection [19, 5].

Sparse principal component analysis (SPCA) addresses this limitation by encouraging principal components to depend on only a small subset of variables. Over the past two decades, SPCA has been studied through several complementary formulations, including regression-based relaxations, power-method variants, and nonsmooth constrained optimization approaches [19, 11, 5]. More recent work has emphasized the importance of handling sparsity and orthogonality in a principled manner, rather than relying solely on heuristic thresholding or post hoc orthogonalization. In particular, manifold-based formulations such as alternating manifold proximal-gradient methods show that sparse multi-component estimation can be treated directly as a nonsmooth optimization problem with orthogonality constraints, yielding stronger algorithmic structure and convergence guarantees than purely heuristic alternatives [5, 4].

Despite these advances, sparsity alone is often not enough. In many high-dimensional domains, features are not exchangeable but are linked through known relational structure. Genes participate in pathways and interaction networks, voxels exhibit spatial or functional connectivity, and pixels or spectral bands obey spatial adjacency and smoothness patterns. When such prior structure is available, it is natural to ask whether the extracted principal components should also respect it. A large literature on graph-aware and graph-regularized PCA studies how incorporating structural priors through graph-based penalties or low-frequency graph constraints can improve robustness and structural fidelity, especially when the signal is aligned with the underlying graph [2]. At the same time, this literature also makes clear that graph-aware PCA is not a single unified model class: some methods emphasize low-frequency graph consistency, some include explicit sparsification, and others target application-specific clustering or representation objectives rather than sparse orthogonal component extraction [2, 9, 17, 15].

A second challenge concerns the extraction of multiple components. In classical PCA, orthogonality follows directly from eigendecomposition. In sparse settings, however, orthogonality is no longer automatic, and sequential deflation can produce components that are only approximately orthogonal or that degrade cumulatively in quality. This has motivated a continued line of work on simultaneous sparse orthogonal extraction, including recent efforts that explicitly revisit orthogonal SPCA as an active optimization problem rather than a solved issue [5, 1, 6]. These developments suggest that, when multiple interpretable components are required, joint extraction under explicit orthogonality constraints remains preferable to treating sparsity as a purely post-processing step.

There is also a broader methodological backdrop. Beyond SPCA itself, recent work on nonsmooth and structured nonsmooth optimization over Riemannian subman-

ifolds has substantially expanded the toolkit available for nonconvex problems with manifold constraints [12, 4]. Moreover, adjacent problems such as sparse tensor canonical correlation analysis already combine Laplacian regularization, sparse penalties, manifold-based optimization, and modern inner solvers with convergence analysis [18]. Consequently, it is no longer accurate to position graph-coupled nonsmooth manifold optimization as an entirely unexplored conceptual territory. Rather, the more defensible and practically meaningful question is whether these ideas can be brought together in a coherent and effective framework for *PCA proper*, with an explicit focus on sparse multi-component loading estimation under feature-graph smoothness and simultaneous orthogonality.

This paper studies that question. We consider a formulation that combines three desiderata: sparse loadings, smoothness with respect to a known feature graph, and joint orthogonality across multiple extracted components. Our goal is not to claim that graph-aware PCA, graph+sparsity PCA, or nonsmooth manifold optimization are individually new. Instead, we focus on a comparatively underexplored intersection of these themes: *feature-graph Laplacian regularization, sparse multi-component loadings, and explicit simultaneous orthogonality within a modern manifold/proximal optimization framework for PCA*. This positioning is narrower than a broad novelty claim, but it better reflects the current state of the literature and the genuine methodological gap that remains [2, 9, 17, 5].

Concretely, we introduce a graph-smooth sparse orthogonal PCA formulation in which orthogonality is handled through a Stiefel-constrained variable and structured sparse estimation is handled through a graph-regularized Euclidean variable coupled by a quadratic penalty. This split allows us to separate the geometry of orthogonality from the graph-coupled sparsity structure, while still optimizing the two jointly. The resulting viewpoint is particularly attractive in settings where interpretability is not just a matter of selecting a few variables, but of selecting variables whose pattern is coherent with known domain structure.

Our study is guided by two central questions. First, when does feature-graph regularization improve the recovery of interpretable sparse components? Second, what tradeoff does graph regularization induce between explained variance, graph consistency, and computational cost? These questions are important because graph-based regularization can help when the prior graph is informative, but may oversmooth the loadings or reduce variance capture when the graph is misspecified. Accordingly, the paper emphasizes not only model design and optimization, but also the empirical circumstances under which graph priors are beneficial.

Contributions. The main contributions of this work are as follows.

- We study an executed synthetic PCA protocol with four methods only: PCA, A-ManPG, SparseNoGraph, and the Proposed method.
- We restrict the empirical scope to chain and SBM graphs, five random seeds (0–4), and shared evaluation metrics only.
- We distinguish SparseNoGraph as an in-family ablation with $\lambda_2 = 0$ from A-ManPG as the minimal external sparse baseline without graph coupling.

Taken together, these contributions position the paper as a targeted integration of ideas that already exist in neighboring literatures, but whose precise combination for graph-smooth sparse orthogonal PCA remains comparatively underdeveloped. Our aim is therefore not to argue that every ingredient is new, but to show that their combination leads to a useful and well-motivated framework for structured dimensionality reduction in graph-informed high-dimensional settings.

2 Related Work

2.1 Sparse PCA and Orthogonal Component Extraction

Sparse principal component analysis has been studied through regression-based relaxations, truncated power methods, penalized matrix decomposition, semidefinite relaxations, and nonsmooth constrained optimization [19, 16, 8, 11, 13]. A central theme in this literature is that sparse loadings improve interpretability but complicate both optimization and the extraction of multiple components.

A particularly important line of work formulates SPCA directly as a nonsmooth optimization problem over orthogonality-constrained variables. Chen et al. proposed the alternating manifold proximal gradient framework (A-ManPG) for sparse PCA and sparse CCA, showing that sparse loadings and nonconvex orthogonality constraints can be handled within a unified manifold-based optimization scheme with convergence guarantees [5]. More broadly, the ManPG literature establishes a practical optimization template for smooth-plus-nonsmooth objectives on the Stiefel manifold [3, 4].

Orthogonal sparse extraction also remains an active topic rather than a closed one. GeoSPCA emphasizes simultaneous extraction over deflation for orthogonal sparse components [1], and recent work on certifiably optimal orthogonal SPCA further underscores ongoing interest in jointly enforcing sparsity, orthogonality, and strong optimality properties [6]. These developments motivate explicit joint orthogonality rather than treating it as an afterthought.

2.2 Graph-Aware PCA and Graph-Regularized Dimensionality Reduction

A parallel literature incorporates structural priors into PCA through graphs or Laplacian-type regularization. These methods are motivated by the observation that in many high-dimensional applications, features are not exchangeable but related through biological, spatial, or network structure. Graph-aware PCA methods use this prior information to bias principal components toward graph-consistent directions, often by suppressing high-frequency variation and preserving graph-coherent low-frequency signals [2, 7, 14].

Recent work makes it clear that graph-regularized PCA cannot simply be characterized as producing dense loadings. In particular, GR-PCA incorporates a learned sparse precision graph and regularizes PCA toward the low-frequency Fourier modes of the induced graph Laplacian; the method also evaluates sparsity-related behavior explicitly and reports improved structural fidelity, lower Laplacian energy, and competitive out-of-sample reconstruction under graph-aligned signals [2]. This weakens any broad claim that graph-aware PCA is predominantly dense or uninterpretable.

At the same time, graph-aware PCA does not fully eliminate the gap relevant here. Many graph-regularized variants focus primarily on graph consistency, robustness, clustering structure, or low-frequency filtering, rather than on simultaneous sparse multi-component orthogonality in a single PCA-specific framework. Accordingly, the relevant distinction is no longer whether graph-aware PCA exists, but whether it addresses the specific combination of sparse loadings, feature-graph smoothness, and explicit joint orthogonality considered here.

2.3 Graph + Sparsity PCA Variants

Older and application-driven papers already show that graph structure and sparsity have been combined within PCA-like models. For example, gLSPCA integrates a graph Laplacian term and a sparsity constraint into PCA for gene-expression analysis [9], while GDSPCA combines graph Laplacian regularization with double sparsity constraints to improve interpretability and exploit internal geometry [17]. These papers are direct counterexamples to any claim that graph+sparsity PCA is absent from the literature.

However, these graph+sparsity PCA methods also illustrate why the gap must be phrased carefully. They are often application-specific, may target particular geometric or clustering objectives, and do not necessarily adopt the same optimization language as modern manifold proximal methods for joint orthogonal extraction. Recent applied work such as RSSPCA reinforces the same point: smooth sparse PCA variants continue to emerge in domain-specific settings, but they do not by themselves close the gap for feature-graph Laplacian regularization with explicit simultaneous orthogonal extraction in PCA proper [15].

2.4 Structured Sparse PCA

Structured sparse PCA extends plain ℓ_1-based sparsity by encouraging supports that follow meaningful patterns such as groups, trees, or contiguous regions. Jenatton, Obozinski, and Bach showed that structured regularization can impose higher-order information on the support of loading vectors and can improve interpretability relative to unstructured sparse PCA [10]. This literature is highly relevant because it demonstrates that sparsity alone is often too weak a prior.

Nevertheless, structured sparse PCA in this sense is not identical to feature-graph smoothness. Structured sparsity typically constrains the support geometry through groups or shapes, whereas graph-Laplacian regularization encourages neighboring or strongly connected features to take similar values. These are related but distinct inductive biases, and the existence of structured sparse PCA does not by itself resolve the specific question of how to jointly impose graph smoothness, sparsity, and simultaneous orthogonality in multi-component PCA.

2.5 Structured Nonsmooth Optimization on Manifolds

The optimization backdrop for this problem has evolved substantially. Beyond the sparse-PCA-specific literature, recent work on nonsmooth and structured nonsmooth optimization over Riemannian submanifolds shows that manifold-based methods now handle a broader class of composite objectives than earlier sparse-only formulations [12, 4]. Recent adjacent work also combines sparse regularization, Laplacian regularization, manifold proximal updates, semismooth Newton subproblem solvers, and convergence analysis in related problems such as sparse tensor CCA [18]. That literature does not solve our PCA formulation directly, but it does weaken any broad claim that the optimization ingredients themselves are unprecedented.

2.6 Positioning of the Present Work

Graph-aware PCA and graph-regularized sparse PCA have been studied in multiple forms, and we do not claim novelty in combining graph smoothness with sparsity. The specific gap we target is the joint extraction of multiple explicitly orthogonal components under a split-variable formulation with a concrete, reproducible solver. Our work is positioned as a focused intersection of sparse PCA, graph-regularized PCA, and orthogonality-preserving optimization, rather than as a first instance of any single ingredient.

3 Problem Setup

Let $X \in \mathbb{R}^{n\times p}$ be a centered data matrix and let

$$\widehat{\Sigma} = \frac{1}{n} X^\top X \tag{1}$$

denote the sample covariance matrix. Let $G = (\mathcal{V}, \mathcal{E}, W)$ be a weighted undirected feature graph on the p variables, with adjacency matrix W, degree matrix D, and graph Laplacian

$$L = D - W. \tag{2}$$

We assume $L \succeq 0$. We use $\|M\|_1$ for the entrywise ℓ_1 norm, $\mathrm{soft}(Z, \tau)$ for entrywise soft-thresholding at threshold τ, and $\mathrm{sym}(M) = (M + M^\top)/2$ for the symmetric part of a matrix. Shared explained variance is computed after QR-based orthonormalization of the loading matrix.

Our goal is to extract r principal components with loading structure that is simultaneously sparse, graph-smooth, and orthogonal. To separate the geometry of orthogonality from the structured regularization, we introduce two variables $A, B \in \mathbb{R}^{p\times r}$:

- A carries the orthogonality constraint $A^\top A = I_r$,
- B carries sparsity and graph smoothness.

We denote the Stiefel manifold by

$$\mathrm{St}(p, r) = \{A \in \mathbb{R}^{p\times r} : A^\top A = I_r\}. \tag{3}$$

4 Proposed Formulation

We study the split formulation

$$\begin{aligned} \min_{A\in\mathrm{St}(p,r),\, B\in\mathbb{R}^{p\times r}} \quad & -\operatorname{tr}(A^\top \widehat{\Sigma} A) + \lambda_1 \|B\|_1 \\ & + \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2. \end{aligned} \tag{4}$$

The first term promotes variance capture; the ℓ_1 term promotes sparse loadings; the Laplacian term promotes graph smoothness; and the quadratic coupling term forces the orthogonal variable A and the structured loading variable B to remain close.

Why split variables? Directly optimizing a sparse graph-regularized objective under orthogonality constraints is difficult because the orthogonality constraint is nonconvex while the graph-sparsity regularizer is nonsmooth and coupled across features. The split formulation isolates these effects: A is updated on the Stiefel manifold, while B is updated in Euclidean space under structured regularization.

We defer alternative sparsity structures to later work to keep the paper-1 formulation fixed to entrywise ℓ_1 on B.

5 Optimization Algorithm

We solve the split-variable objective by alternating between two steps. The A-step updates an orthogonal loading matrix on the Stiefel manifold using a Riemannian gradient step followed by QR-based retraction. The B-step updates the sparse graph-smooth proxy with proximal gradient, treating the Laplacian quadratic and coupling term as the smooth part and the ℓ_1 penalty as the nonsmooth part. This single solver path is the primary algorithmic target for paper 1; more advanced variants are deferred.

5.1 A-Step: Manifold Update

With B fixed, the A-subproblem is

$$\min_{A\in\mathrm{St}(p,r)} -\operatorname{tr}(A^\top \widehat{\Sigma} A) + \frac{\rho}{2}\|A - B\|_F^2. \tag{5}$$

Its Euclidean gradient is

$$\nabla_A f(A; B) = -2\widehat{\Sigma} A + \rho(A - B). \tag{6}$$

The corresponding Riemannian gradient on $\mathrm{St}(p, r)$ is

$$\operatorname{grad} f(A; B) = \nabla_A f(A; B) - A \operatorname{sym}\big(A^\top \nabla_A f(A; B)\big). \tag{7}$$

We take a fixed-step manifold descent update with step size η_A and retract back to the Stiefel manifold using QR retraction.

5.2 B-Step: Graph-Coupled Sparse Update

With A fixed, the B-subproblem is

$$\min_{B\in\mathbb{R}^{p\times r}} \lambda_1 \|B\|_1 + \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2. \tag{8}$$

This subproblem is convex in B because $L \succeq 0$. We use a proximal-gradient update that treats the Laplacian quadratic and coupling term as the smooth part and the ℓ_1 penalty as the nonsmooth part. Define

$$g(B) = \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2, \qquad h(B) = \lambda_1 \|B\|_1.$$

Then

$$\nabla g(B) = 2\lambda_2 L B + \rho(B - A).$$

With step size $\eta_B = 1/(2\lambda_2 \|L\|_2 + \rho)$, the update is

$$B^{(t+1)} = \mathrm{soft}\big(B^{(t)} - \eta_B \nabla g(B^{(t)}), \eta_B \lambda_1\big).$$

5.3 Alternating Scheme

Algorithm 1 summarizes the outer loop.

Algorithm 1 Alternating Manifold + Graph-Coupled Sparse Updates

Require: Centered data X, graph Laplacian L, parameters $\lambda_1, \lambda_2, \rho$, rank r

Initialize $A^{(0)} \in \mathrm{St}(p, r)$, $B^{(0)} \in \mathbb{R}^{p \times r}$
for $t = 0, 1, \ldots, T-1$ **do**
  Update $A^{(t+1)}$ with one Riemannian gradient step and QR retraction
  Update $B^{(t+1)}$ with one proximal-gradient step and soft-thresholding
  if stopping criterion satisfied **then**
    break
  end if
end for
return $(A^{(T)}, B^{(T)})$

Complexity. The cost of the A-step is dominated by matrix multiplications and the chosen retraction. The B-step is typically the computational bottleneck because graph regularization introduces feature coupling and removes the closed-form separability available in plain soft-thresholding. When L is sparse, matrix-vector operations remain efficient; dense or ill-conditioned graph operators can substantially increase runtime.

6 Optimization Discussion

The objective in (4) is nonconvex because it combines a Stiefel constraint on A with a quadratic coupling term between A and B. For fixed A, however, the B-subproblem is convex because it is the sum of a convex quadratic term and the entrywise ℓ_1 norm.

6.1 Updates Used

The implemented solver alternates between two explicit updates. The A-step takes one retracted Riemannian gradient step on $\mathrm{St}(p, r)$ using the Euclidean gradient from the manifold objective and QR retraction back to the manifold. The B-step takes one proximal-gradient step, with the smooth part given by the Laplacian quadratic and coupling term and the nonsmooth part given by $\lambda_1 \|B\|_1$.

The step size for the A-step is fixed at the experimental value $\eta_A = 0.05$. The B-step uses

$$\eta_B = \frac{1}{2\lambda_2 \|L\|_2 + \rho},$$

which matches the Lipschitz constant of the smooth part in the Euclidean update.

6.2 Stopping Criteria

The outer loop stops when any of the following is satisfied:

- relative objective change falls below a tolerance,
- the coupling gap $\|A - B\|_F$ falls below a tolerance,
- the orthogonality defect $\|A^\top A - I_r\|_F$ falls below a tolerance,
- the maximum iteration count is reached.

This section is intentionally descriptive rather than a proof development: we do not provide a formal convergence proof here, and formal analysis is out of scope for this manuscript.

7 Experimental Setup

We evaluate only the executed synthetic study. The comparison set is limited to PCA, A-ManPG, SparseNoGraph, and the Proposed method. SparseNoGraph is the in-family ablation with $\lambda_2 = 0$, while A-ManPG is the minimal external sparse orthogonal baseline without graph coupling. The graph families are chain and SBM, and all main results use seeds 0–4. Shared evaluation metrics only are reported: support recovery, graph smoothness on the true graph, and shared explained variance. The explained-variance quantity is an evaluation metric computed after QR-based orthonormalization, not a training objective for any method.

7.1 Main Runs

The main runs use the following hyperparameters unless otherwise stated:

- $\lambda_1 = 0.1$,
- $\lambda_2 = 0.1$,
- $\rho = 5.0$,
- $\eta_A = 0.05$.

7.2 SBM Sweep

We also report a single-seed SBM sweep over λ_2 using the same method comparison set. The sweep is used to illustrate the recovery–smoothness tradeoff under increasing graph regularization.

8 Results

Overview. We report synthetic results using shared evaluation metrics across all methods: union support F1, graph smoothness norm on the true graph, and a shared explained-variance measure computed from QR-orthonormalized loadings. Shared explained variance is an evaluation metric and is not the training objective for any of the methods. We do not compare raw objective values across methods because they optimize different objectives. Results are presented for chain and SBM graphs across five seeds (0–4), and for a single-seed SBM

Table 1: Chain robustness (seeds 0–4). Mean ± std for shared evaluation metrics.

Method	Support F1	Smoothness (norm)	Shared explained var
PCA	0.261 ± 0.000	2.099 ± 0.484	6.225 ± 0.406
A-ManPG	0.914 ± 0.078	2.111 ± 0.483	6.122 ± 0.408
SparseNoGraph	0.950 ± 0.033	2.118 ± 0.476	6.092 ± 0.413
Proposed	0.927 ± 0.019	1.871 ± 0.465	6.059 ± 0.415

Table 2: SBM robustness (seeds 0–4). Mean ± std for shared evaluation metrics.

Method	Support F1	Smoothness (norm)	Shared explained var
PCA	0.261 ± 0.000	17.085 ± 0.974	6.225 ± 0.406
A-ManPG	0.914 ± 0.078	17.134 ± 1.008	6.122 ± 0.408
SparseNoGraph	0.950 ± 0.033	17.166 ± 1.035	6.092 ± 0.413
Proposed	0.744 ± 0.051	15.764 ± 1.104	6.003 ± 0.384

sweep over λ_2. The comparison set includes PCA, A-ManPG, SparseNoGraph, and the Proposed method.

8.1 Chain Robustness (Seeds 0–4)

Across seeds, PCA is consistently worst on support recovery. A-ManPG improves substantially over PCA but remains below the sparse baselines on mean support F1. SparseNoGraph achieves the highest mean support F1, while the Proposed method attains slightly lower F1 but achieves a lower graph smoothness norm on the true chain graph. Shared explained variance is similar across the non-PCA methods and lower than PCA, as expected for sparse baselines. This pattern indicates that on chain graphs, graph regularization improves structural coherence while remaining competitive on recovery.

8.2 SBM Robustness (Seeds 0–4)

On SBM graphs, PCA again performs worst on support recovery. A-ManPG again improves over PCA but does not match the sparse baselines on support recovery. SparseNoGraph attains substantially higher support F1 than the Proposed method, while the Proposed method is consistently smoother on the true SBM graph (lower normalized graph smoothness). Shared explained variance remains comparable across the sparse methods. These results show a stable recovery–smoothness tradeoff on SBM: stronger graph regularization improves structural smoothness but can reduce recovery accuracy.

8.3 SBM λ_2 Sweep (Seed 2)

Sweeping $\lambda_2 \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$ reveals an overall smoothness–recovery tradeoff. Light graph regularization improves smoothness with minimal loss in support F1, while larger λ_2 values steadily decrease support F1 and increase support density. Shared explained variance declines moderately as λ_2 grows. This sweep indicates that λ_2 must be tuned to graph topology; values that are benign on chains can oversmooth on SBM.

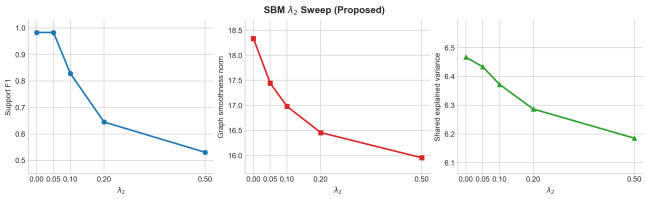

Figure 1: SBM lambda2 sweep (seed 2). Light regularization improves smoothness with limited recovery loss, while stronger regularization oversmooths and degrades support recovery.

9 Discussion and Limitations

These experiments establish a focused empirical message: in our synthetic experiments, graph regularization yields smoother solutions on the true graph for both chain and SBM settings, but its effect on support recovery is topology-dependent. On chain graphs, the proposed method remains competitive on recovery while producing smoother supports, consistent with the intuition that chain structure aligns well with Laplacian smoothness. On SBM graphs, the graph prior enforces smoother structure at the cost of recovery accuracy, yielding a stable tradeoff rather than a contradiction.

The chain setting aligns naturally with local smoothness, whereas the SBM setting emphasizes block/community structure; as a result, the same smoothness penalty can act as a mild regularizer in one topology and as an oversmoothing force in another. The λ_2 sweep on SBM makes this explicit: mild regularization yields smoother solutions with limited recovery loss, whereas stronger regularization oversmooths and reduces recovery. These observations imply that λ_2 must be tuned with respect to graph topology rather than assumed to be universally beneficial.

The main limitation of the current evidence is that it remains synthetic and focuses on two graph families with a single data generator. This is appropriate for Paper 1, whose goal is to establish a disciplined method and a clear empirical tradeoff. The results already demonstrate that structural priors can improve graph coherence, but they also show that regularization strength must be tuned to topology. This is a valid and useful contribution at the current scope.

10 Conclusion

We present synthetic evidence that graph-regularized sparse orthogonal PCA improves structural coherence while inducing a topology-dependent tradeoff with support recovery. On chain graphs, the proposed method is competitive on recovery and yields smoother supports; on SBM graphs, stronger graph regularization improves smoothness but reduces recovery accuracy. These findings highlight the need for topology-aware λ_2 tuning and

support a modest, defensible Paper-1 claim: graph regularization can improve structural coherence, but its effect on recovery depends on graph topology and regularization strength.

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